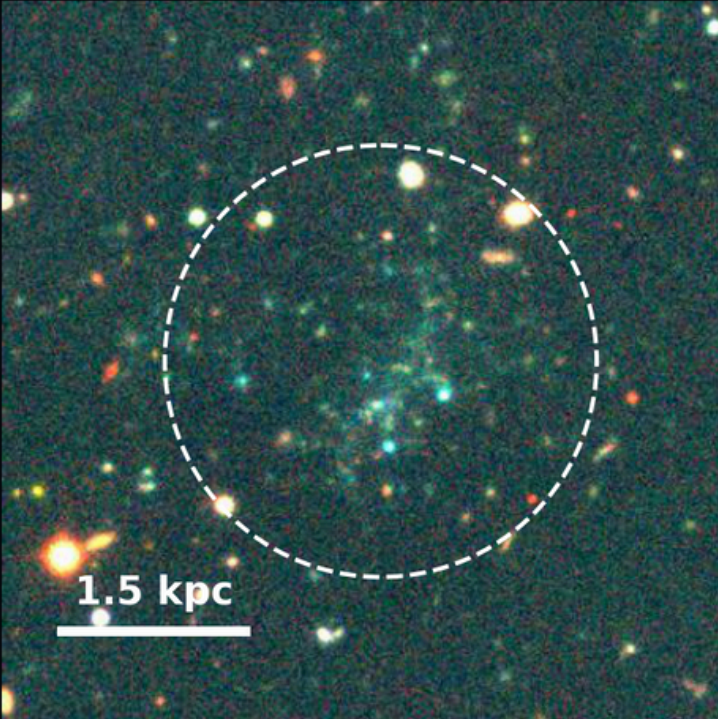


FINDING BLUE BLOBS

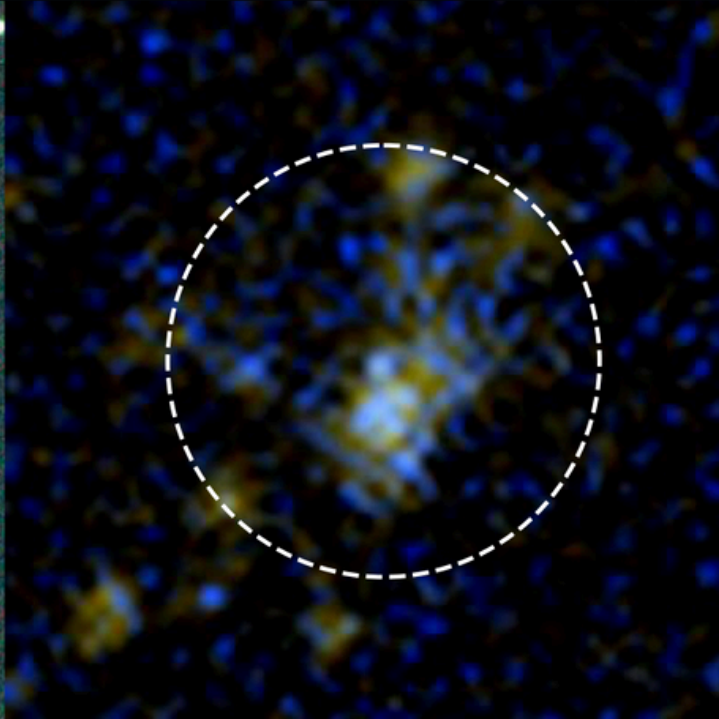
Using citizen science to discover a novel
class of stellar systems.

Swapnaneel Dey

BC1239+1212 a.k.a BC1



Next Generation Vixen Cluster
Survey (NGVS)



Galaxy Evolution
Explorer (GALEX)



HUBBLE SPACE TELESCOPE
(HST)

BACKGROUND

- Blue ($g - i \lesssim 0$)
- Isolated (100s of kpc away from the nearest galaxy)
- Extremely low stellar mass $\left(\log\left(\frac{M_*}{M_\odot}\right) \lesssim 5\right)$

Until recently, only six blue blobs were known.*

*Across these papers:
 E. A. K. Adams et al. (2013)
 M. G. Jones et al. (2022a)
 J. M. Cannon et al. (2015)
 M. G. Jones et al. (2024a)

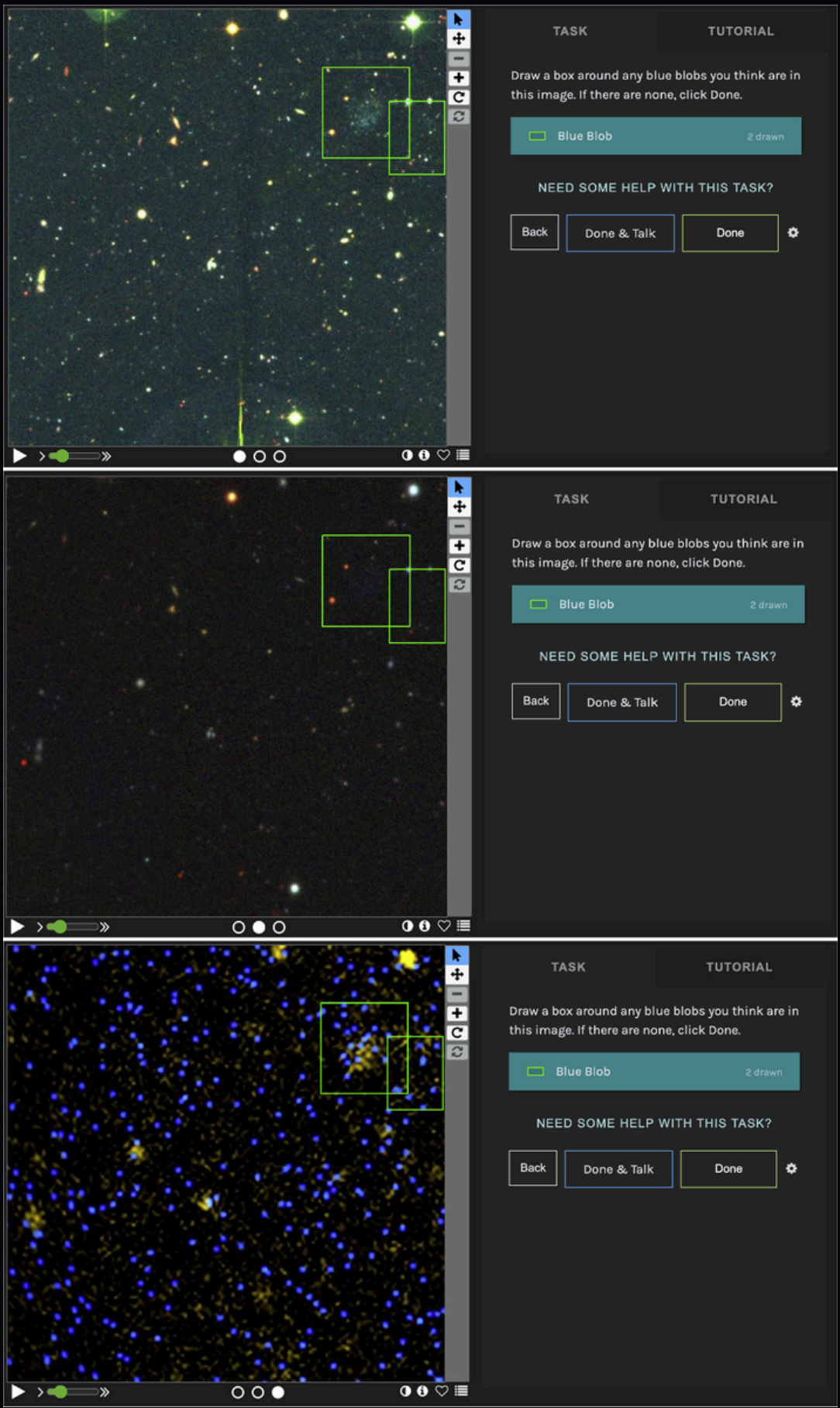
CITIZEN SCIENCE

Launched a Zooniverse project - Blobs and Blurs: Extreme Galaxies in Clusters

- >1600 citizen scientists participated.
- >450,000 classifications made
- 150,000 cutouts were visually searched

658 blue blob candidates classified by **at least 3** volunteers. Out of which **40** were credible blue blobs.

Out of 50,000 images only 40 images had a blue blob.



Zooniverse interface

Volunteers flip through
NGVS, DECaLS, GALEX
image

AIM

Make a model that can differentiate between blue blobs and background sky+galaxies

Problem 1

- Only 40 blue blob images.
- No limit to the amount of background sky+galaxy images.

Problem 2

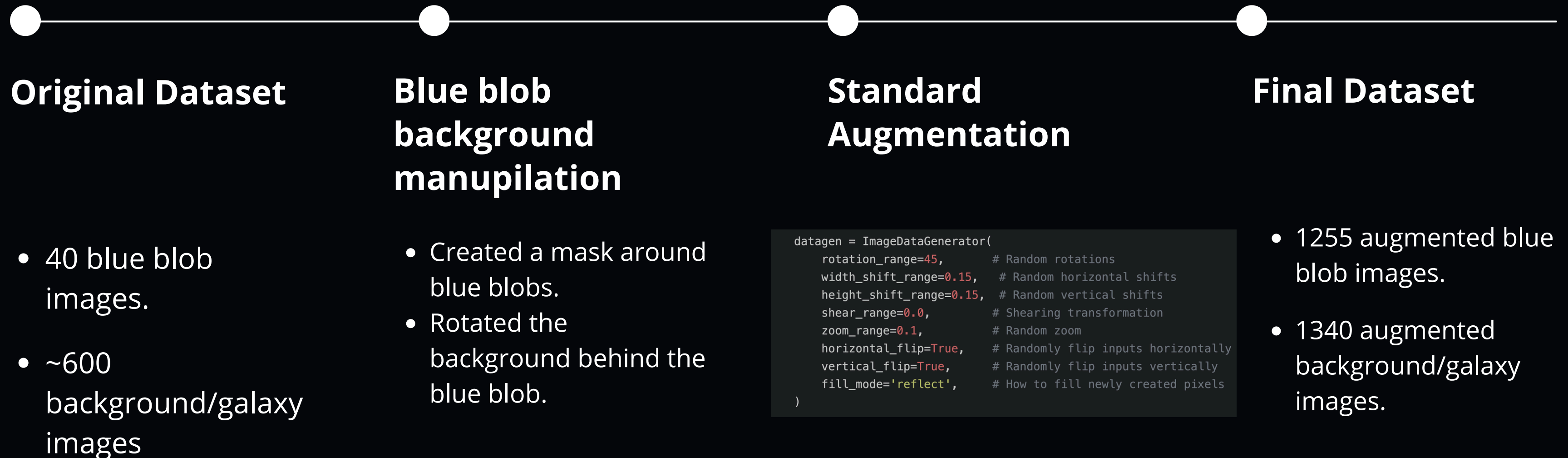
- Can a CNN be trained from scratch?
- Use a pre-trained model, but which one?

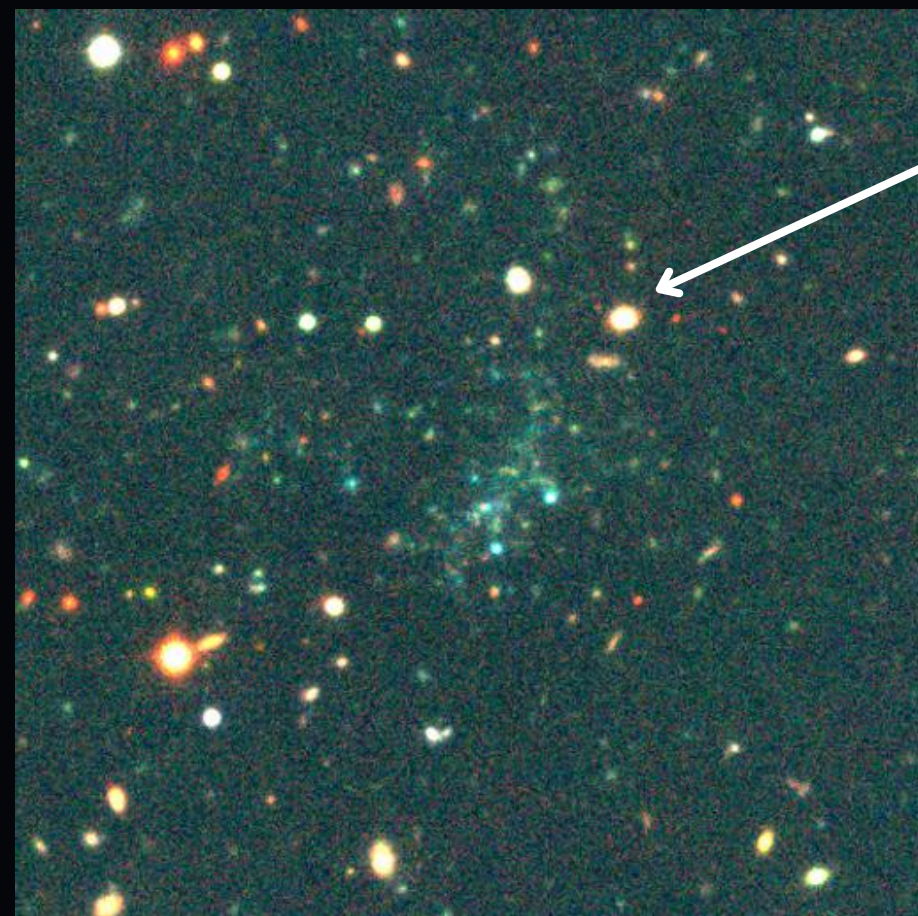
Problem 3

- Will the model ever be good enough to detect such rare objects?

DATA AUGMENTATION

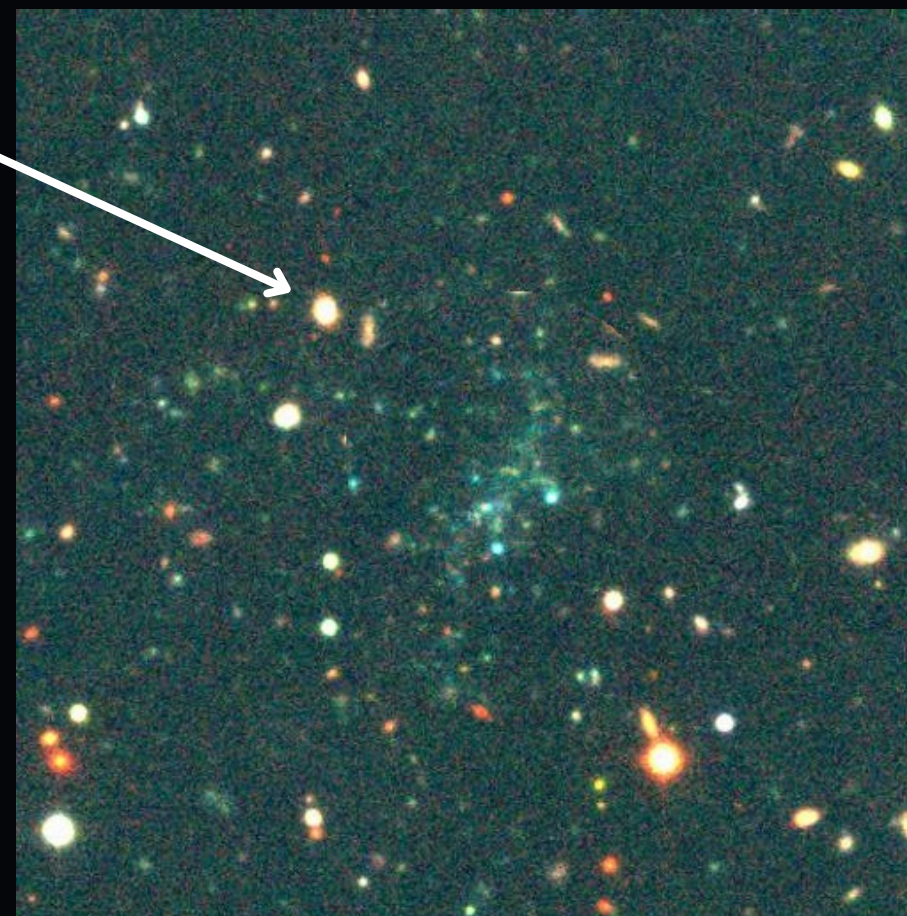
Generating new examples by applying random, label-preserving transformations to your existing data.





Original

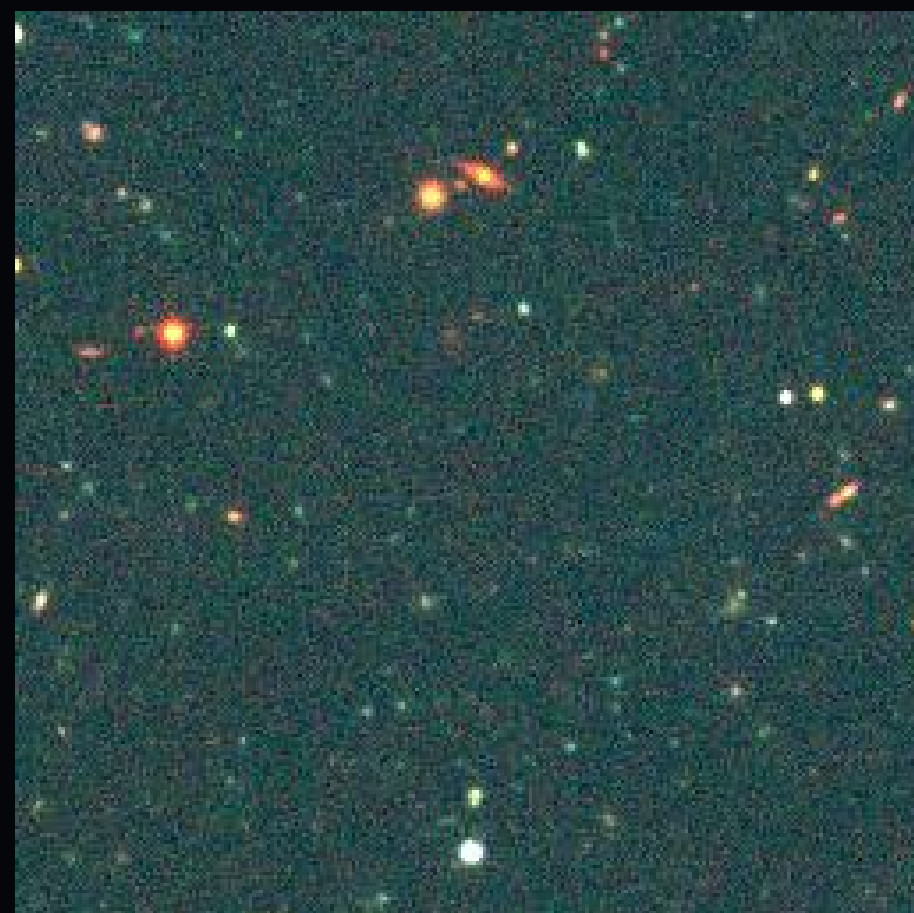
Background
augmentation



Downsampling
and full
augmentation



Final



Full
augmentation



AIM

Make a model that can differentiate between blue blobs and background sky+galaxies

Problem 1

- Only 40 blue blob images.
- No limit to the amount of background sky+galaxy images.

Problem 2

- Can a CNN be trained from scratch?
- Use a pre-trained model, but which one?

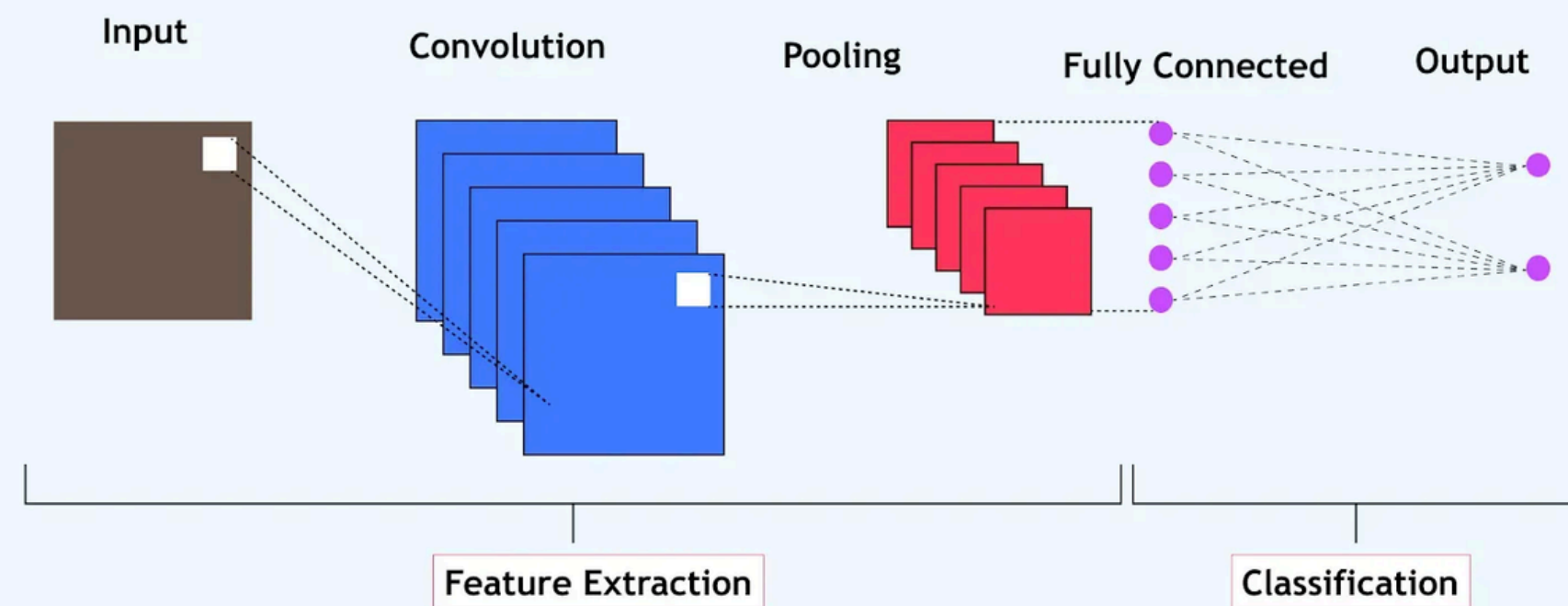
Problem 3

- Should the model ever be trusted if trained on such a small dataset?

TRANSFER LEARNING

- **Transfer Learning:** Use a model trained on one task to jump-start learning on a new, related task.
- **Feature Extraction:** Freeze the pre-trained layers, run your data through them, and train only a small new head.
- **Fine-Tuning:** Unfreeze a few top layers and train jointly with your head to better adapt to your data.
- **Data Efficiency:** Builds on existing patterns—needs less data and cuts down overfitting.

The Architecture of Convolutional Neural Networks



He+ 2016

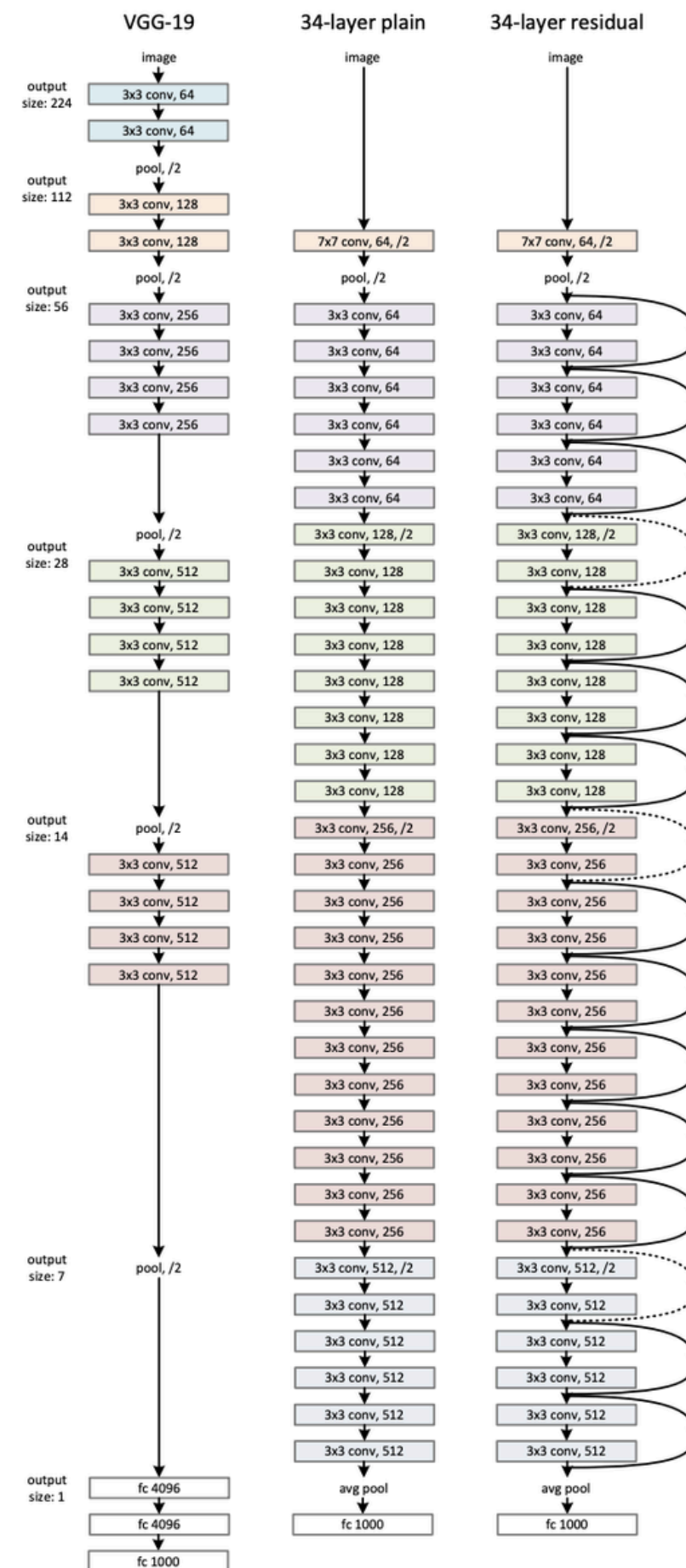


Figure 3. Example network architectures for ImageNet. **Left:** the VGG-19 model [41] (19.6 billion FLOPs) as a reference. **Middle:** a plain network with 34 parameter layers (3.6 billion FLOPs). **Right:** a residual network with 34 parameter layers (3.6 billion FLOPs). The dotted shortcuts increase dimensions. **Table 1** shows more details and other variants.

Comparing different base_models

ResNet50V2

- Parameters: 25.6 million
- Structure: 50 layers deep NN
- High ImageNet accuracy

EfficientNetB3

- Parameters: 11 million
- Structure: 377 layers
- High ImageNet accuracy

ResNet101V2

- Parameters: 44.6 million
- Structure: 101-layer deep
- High ImageNet accuracy

Model Architecture

ResNet50 I

Attached one dense layer with one sigmoid unit.

Fine-tuned by unfreezing the last block.

ResNet50 II

Attached 128-unit ReLU dense layer, then final dense layer with sigmoid activation.

Fine-tuned by unfreezing the last block.

EfficientNetB3

Attached one dense layer with one sigmoid unit.

Fine-tuned by unfreezing the last block.

ResNet101V2

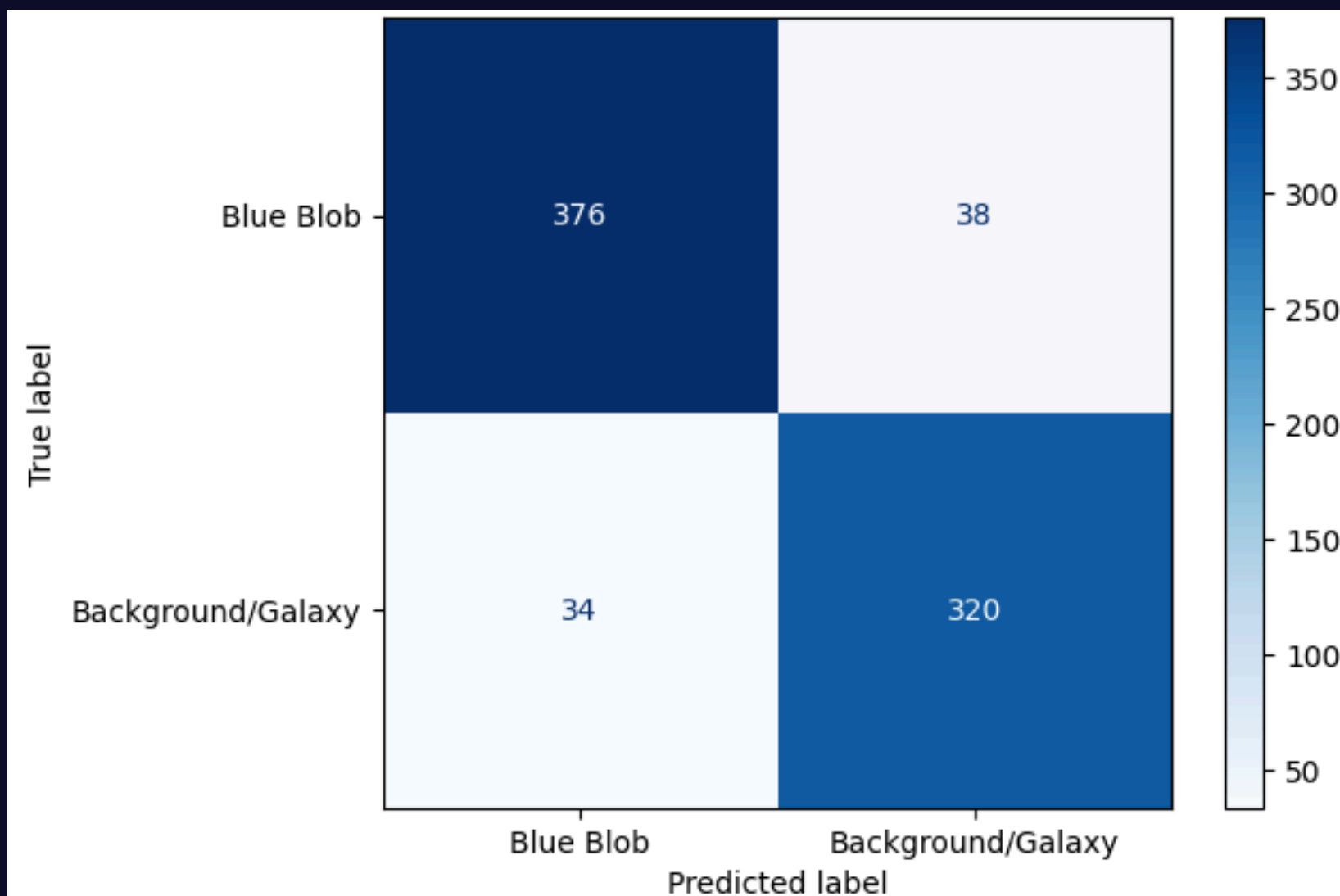
256-unit ReLU dense layer then a 128-unit ReLU dense layer then a final sigmoid output layer.

Fine-tuned by unfreezing the last conv. block.

Training set shape = (1280, 256, 256, 3)
Validation set shape = (547, 256, 256, 3)

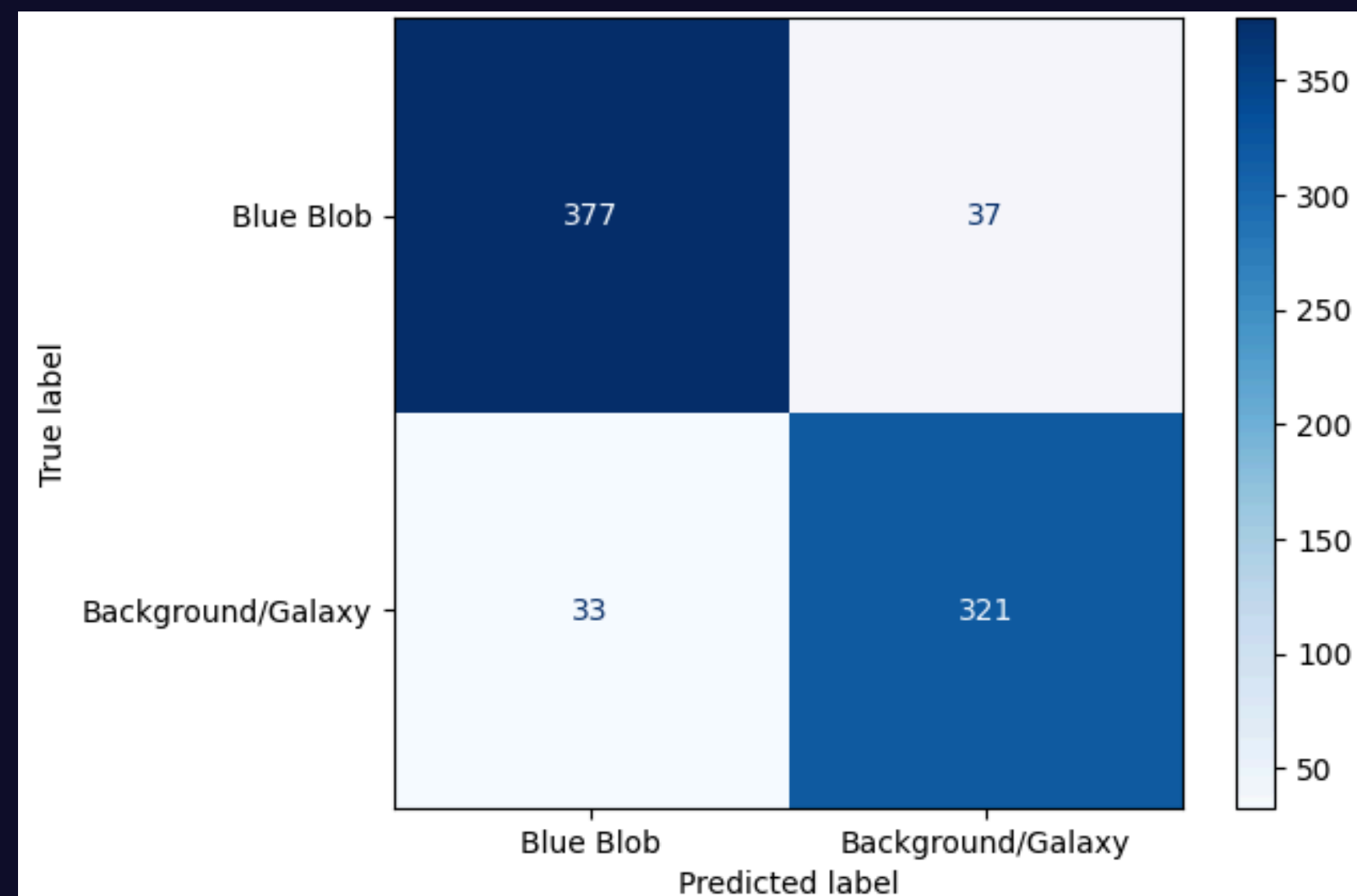
ResNet50 I

New head training



Becomes slightly better
→

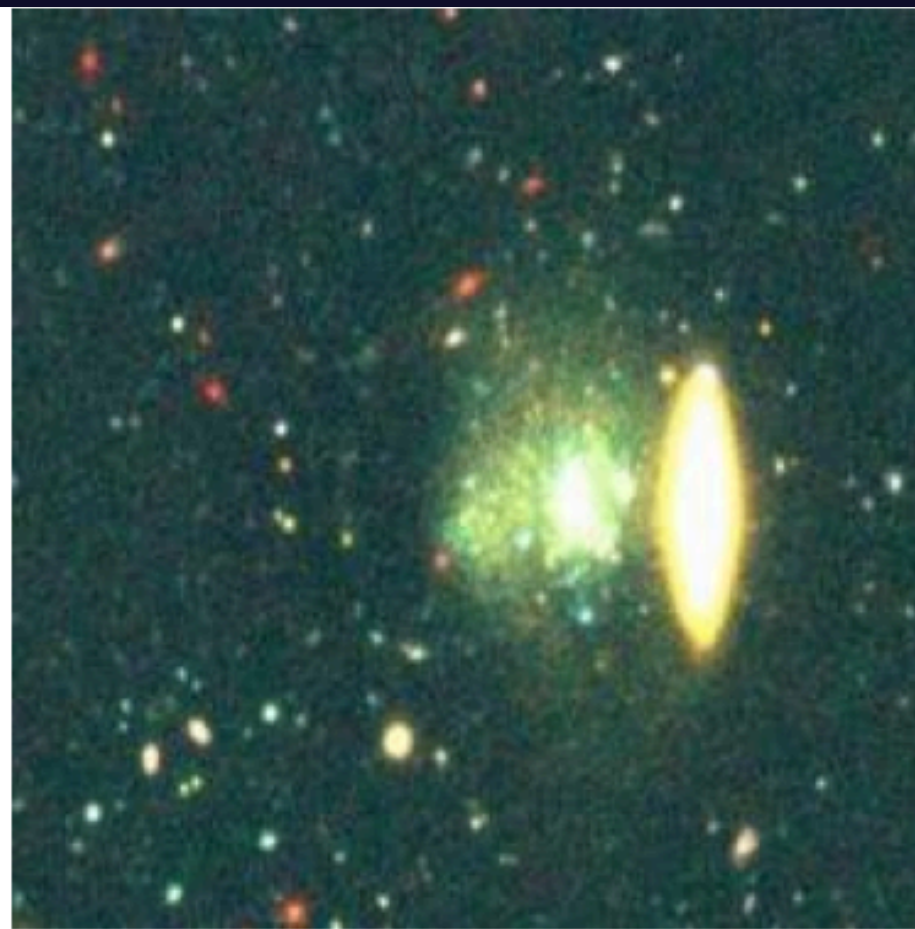
After layer unfreezing fine-tuning phase



Tried to find the best epoch where the validation loss was the minimum.



Some blue blobs
classified as
background or
galaxies.

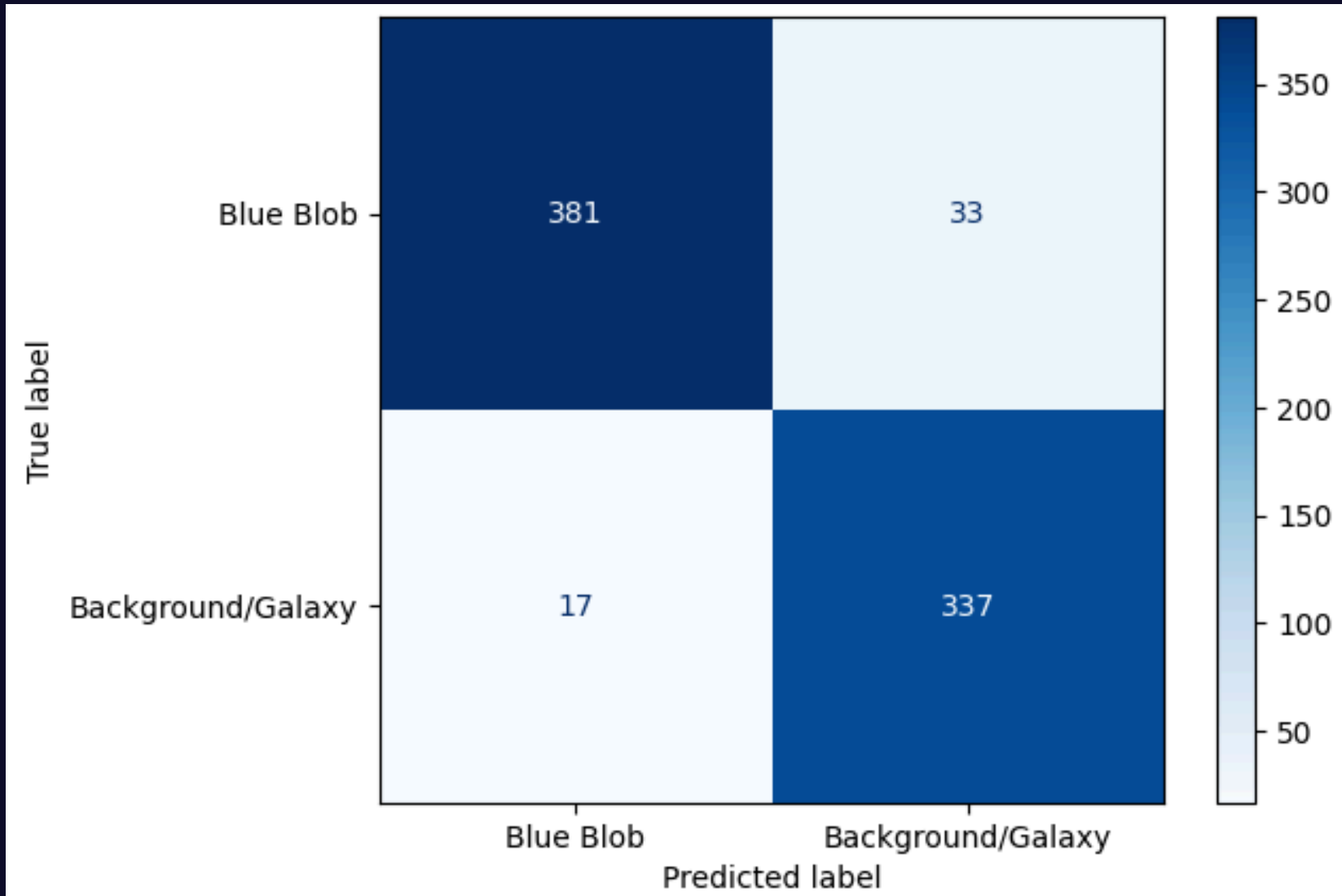


Some background
images classified
as blue blobs



ResNet50 II

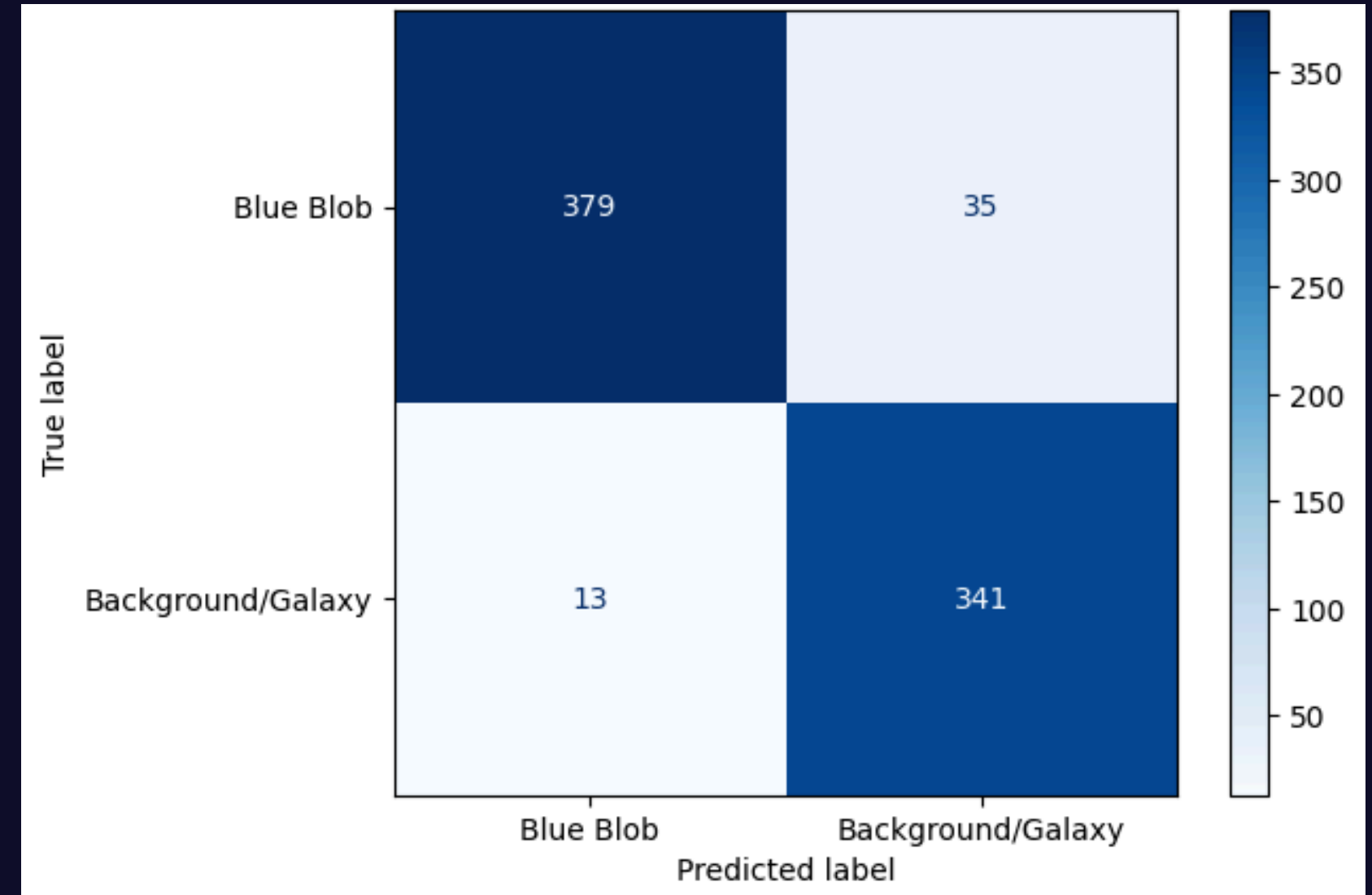
New head training



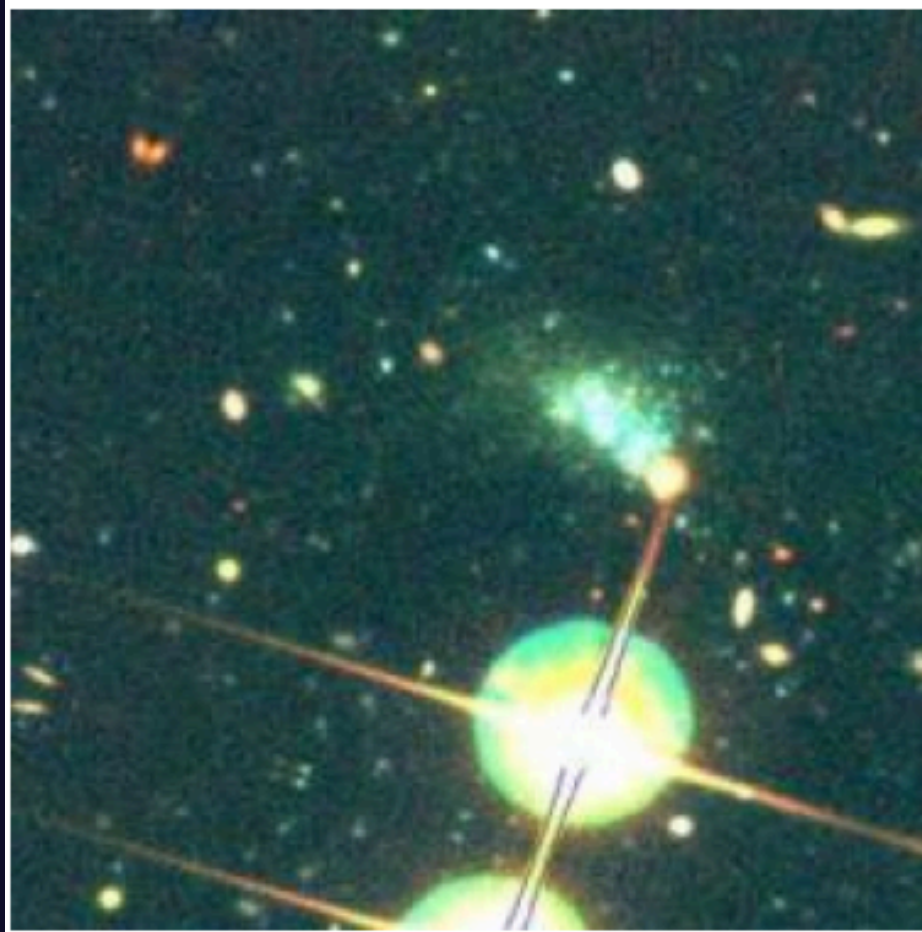
Becomes slightly better



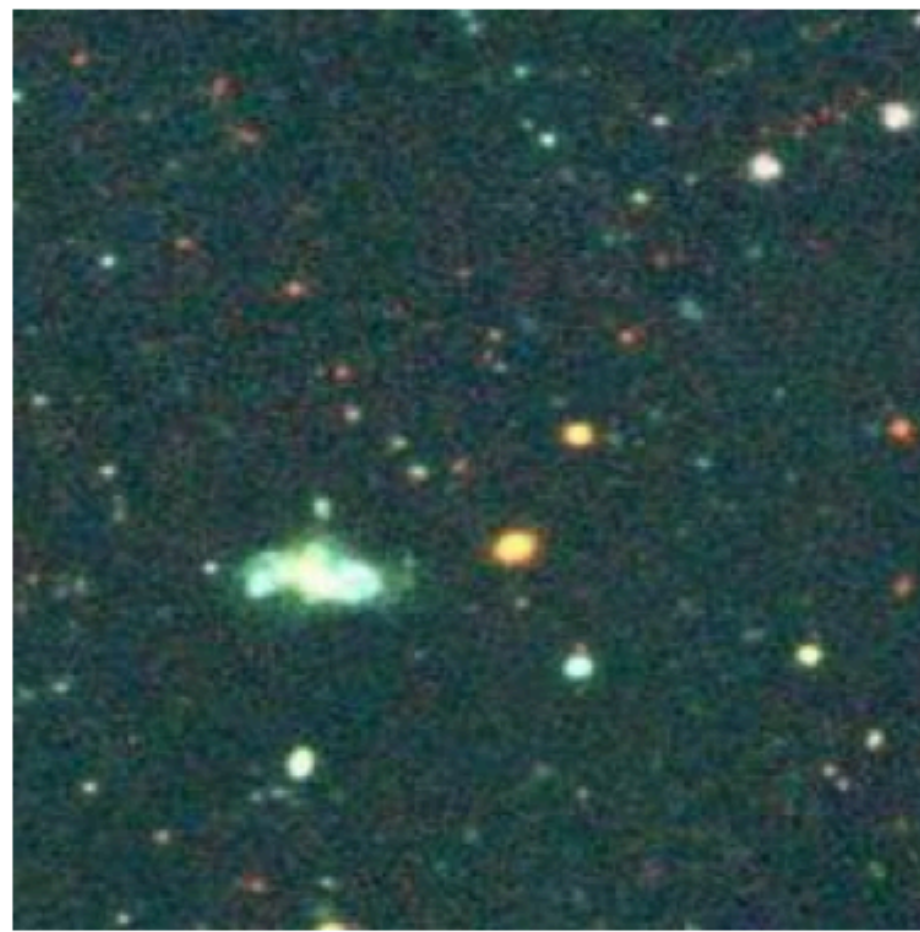
After layer unfreezing fine-tuning phase



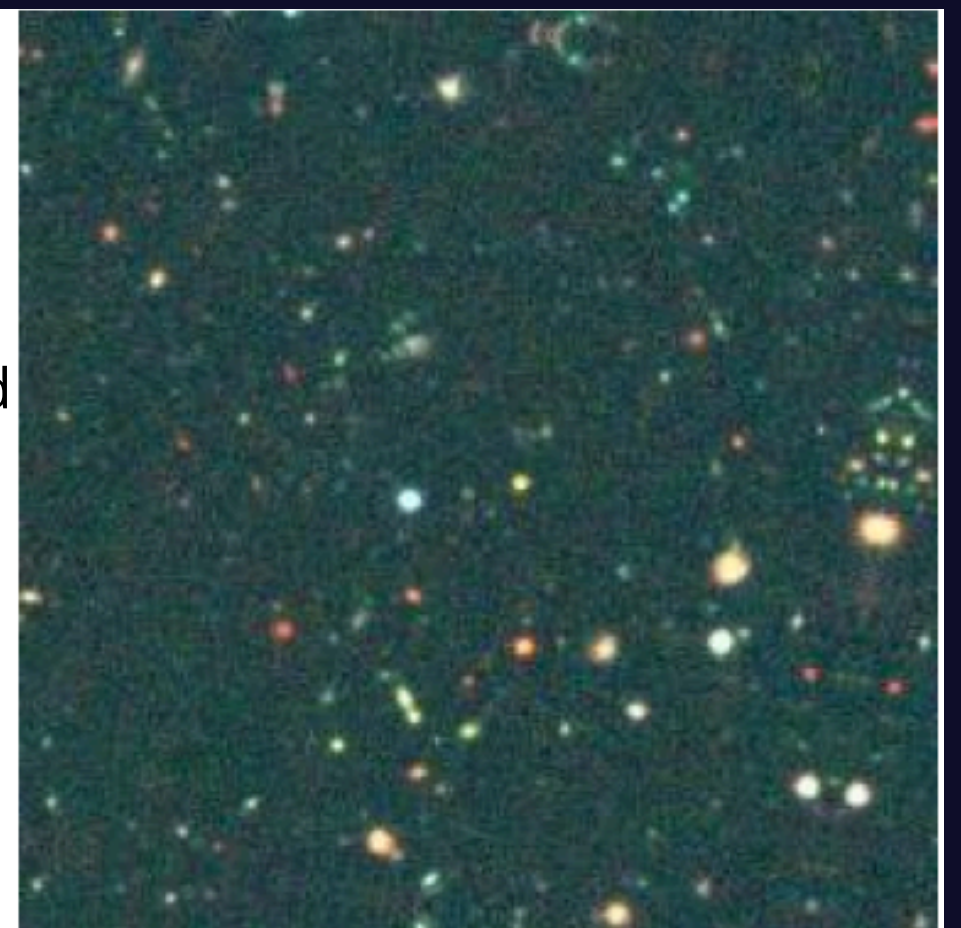
Tried to find the best epoch where the validation loss was the minimum.



Some blue blobs
classified as
background or
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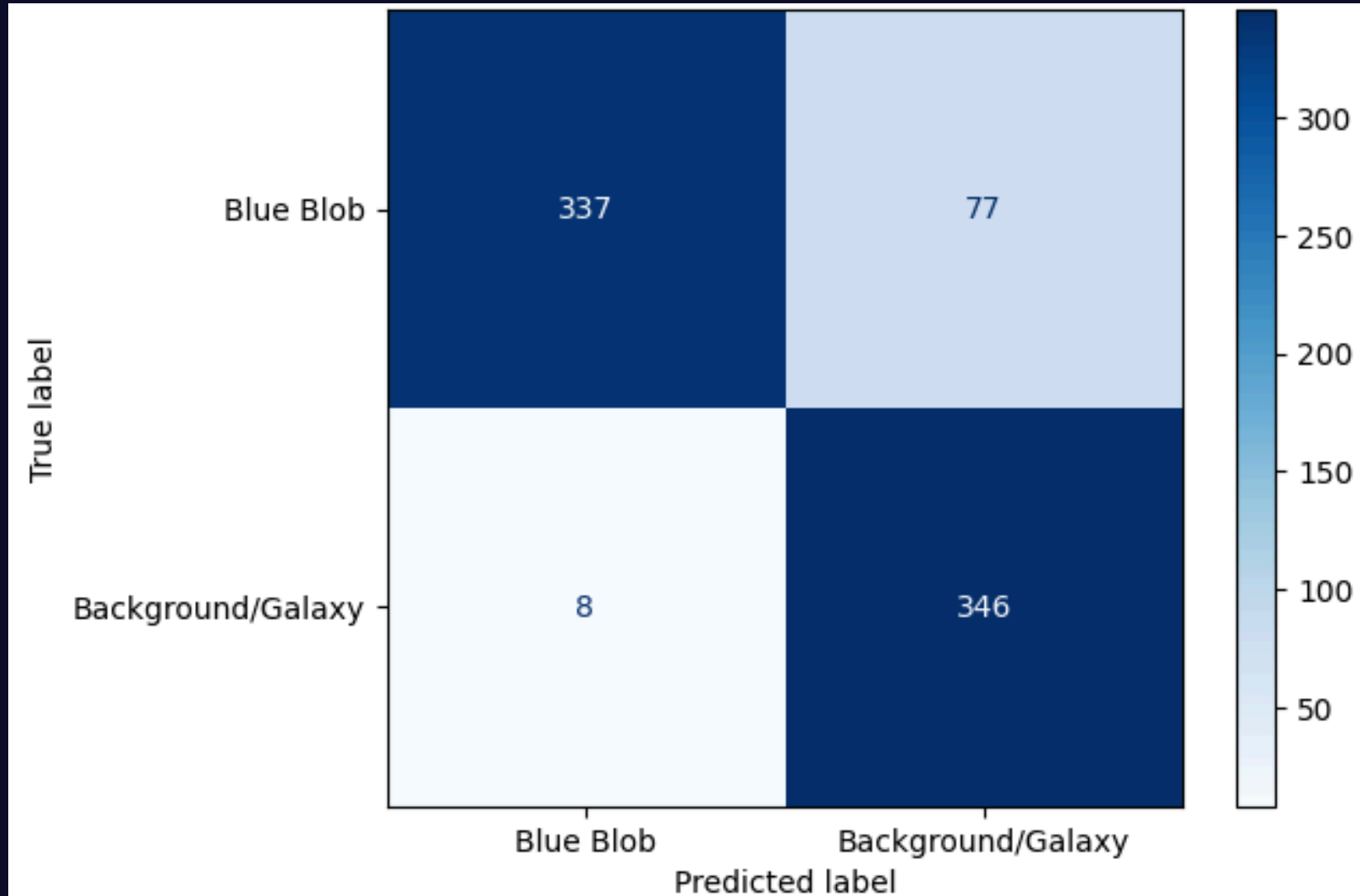


Some background
images classified
as blue blobs



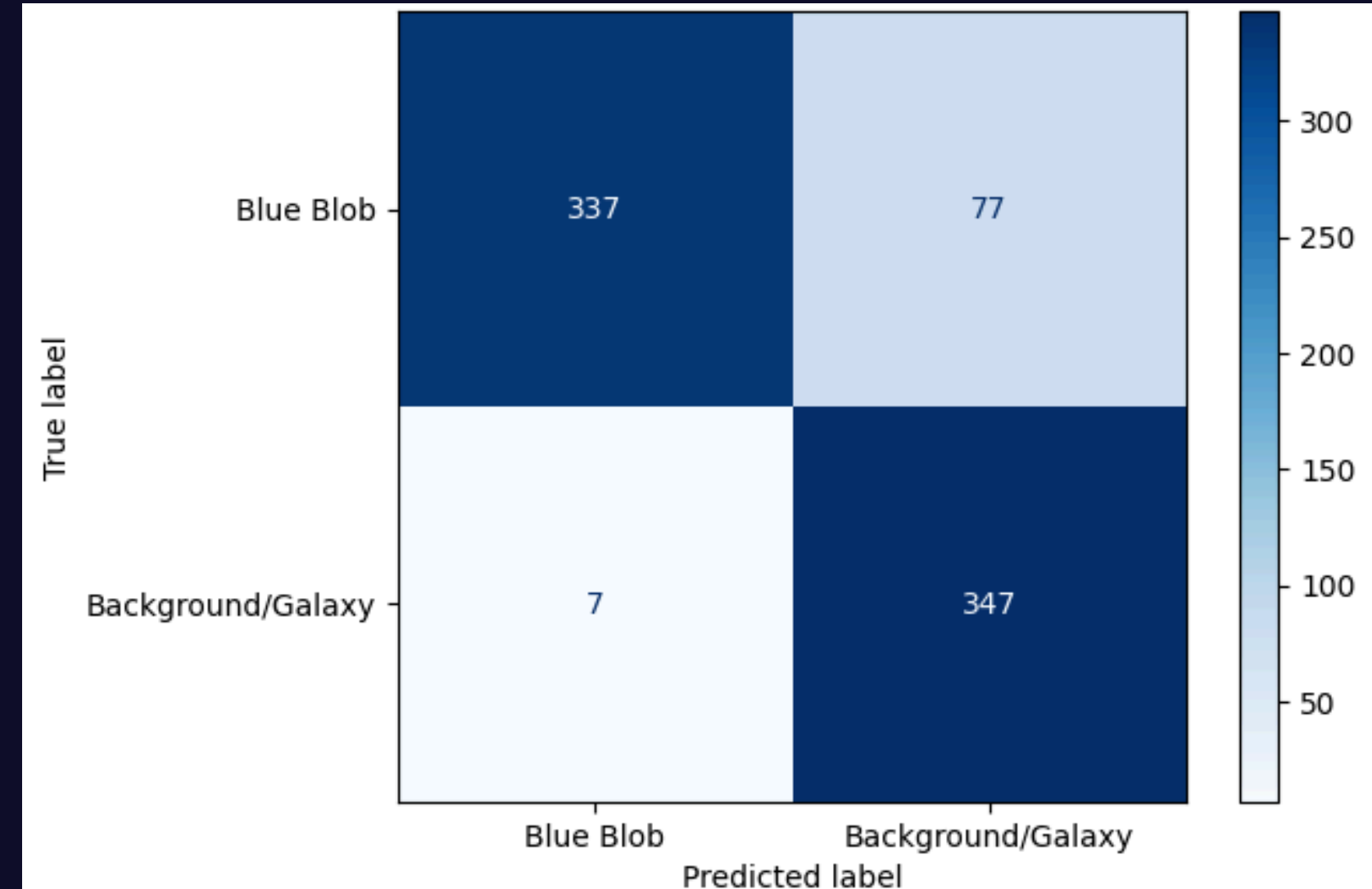
EfficientNetB3

New head training



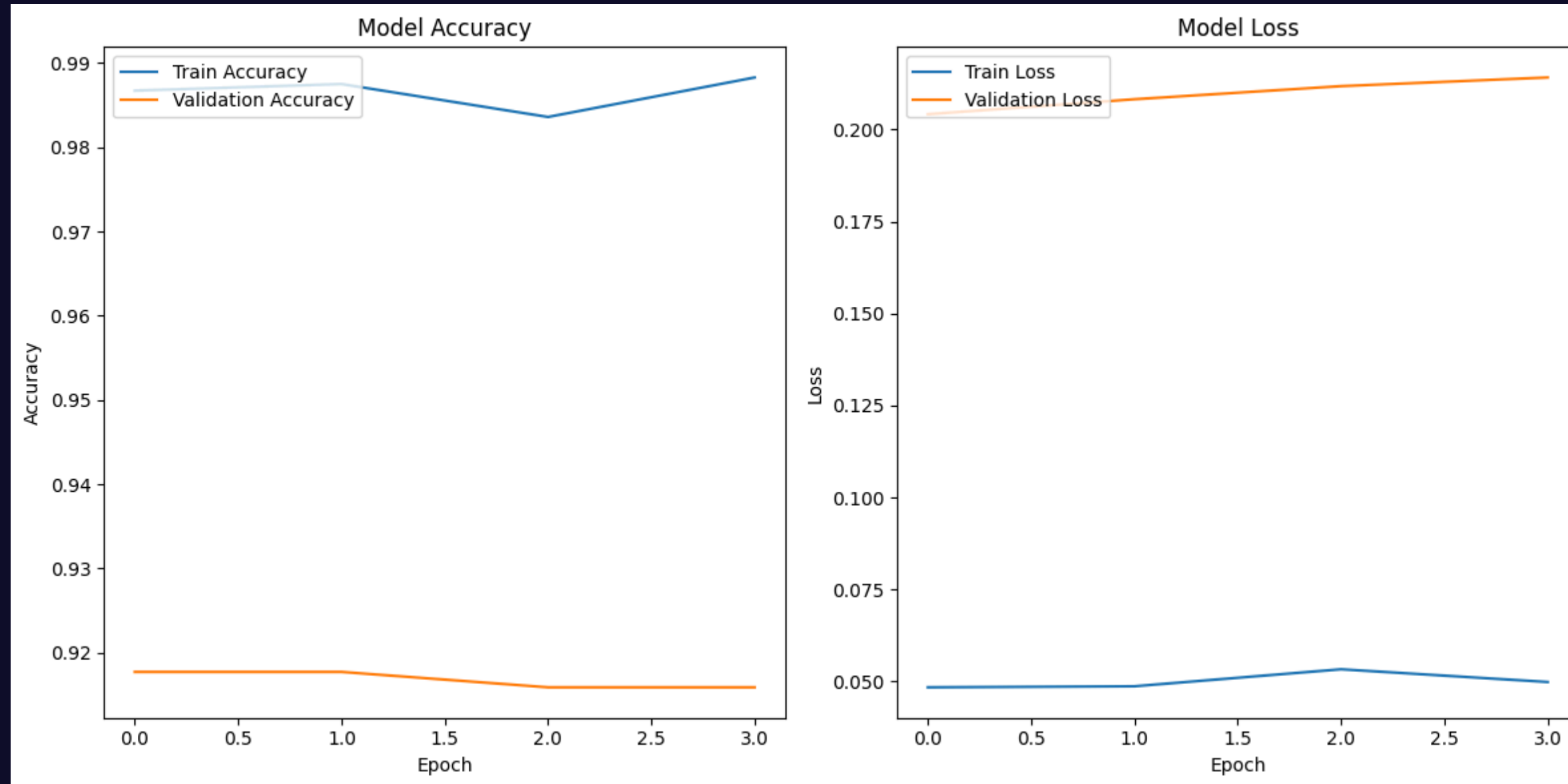
Becomes slightly
better
→

After layer unfreezing fine-tuning phase



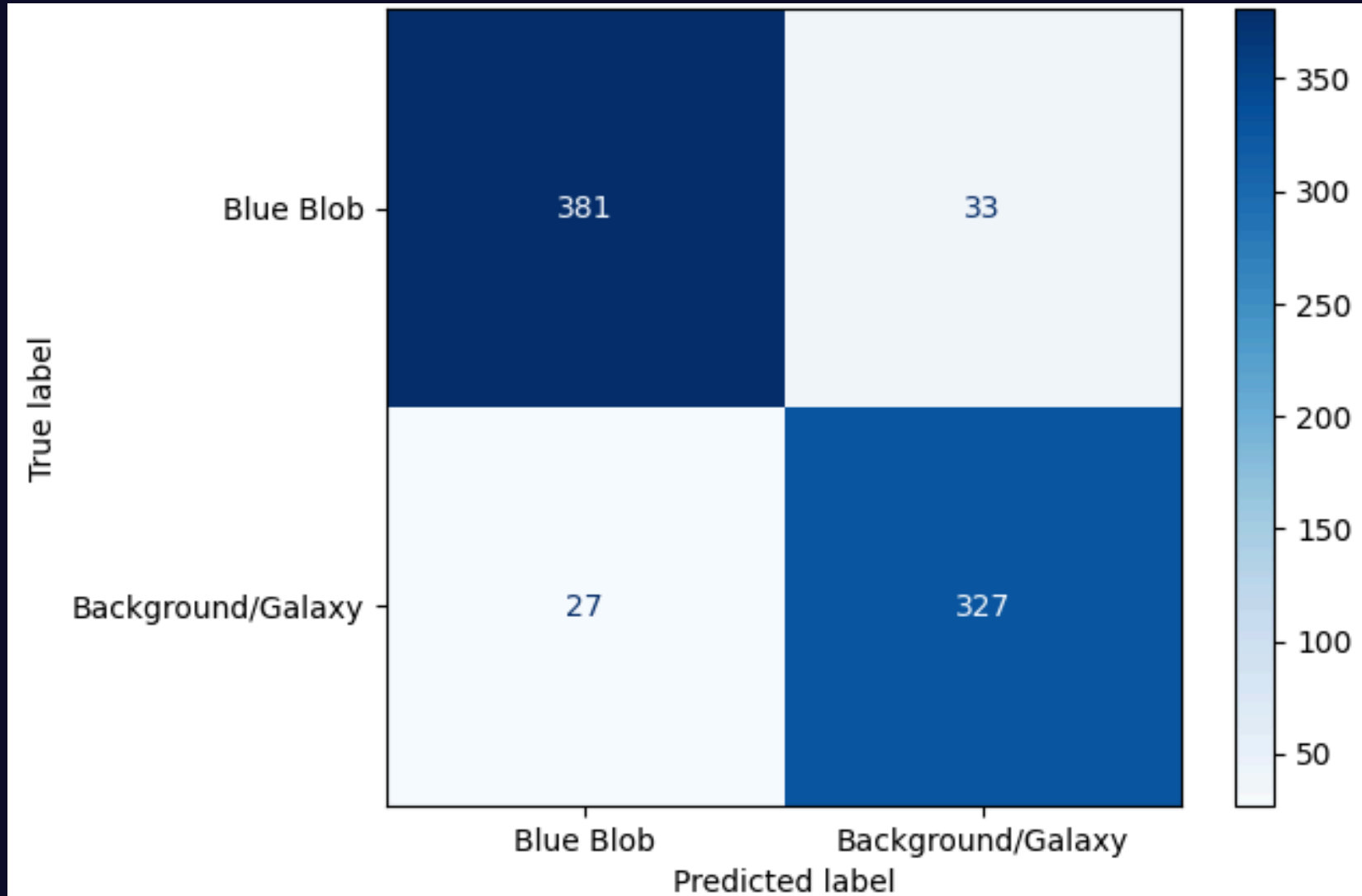
Tried to find the best epoch where the validation loss was the minimum.

A bad fine-tuning phase



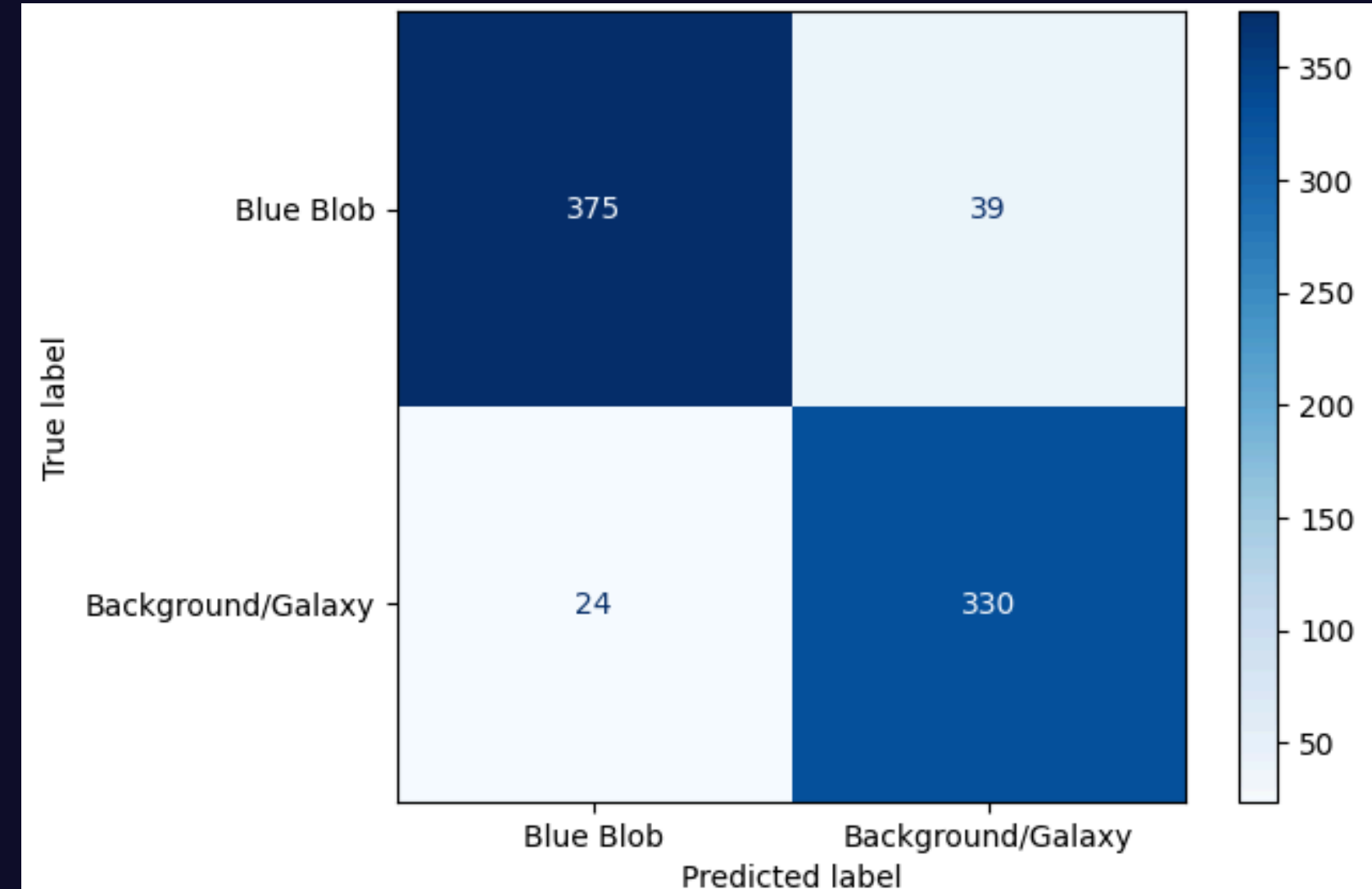
ResNet101V2

New head training



Becomes slightly worse
→

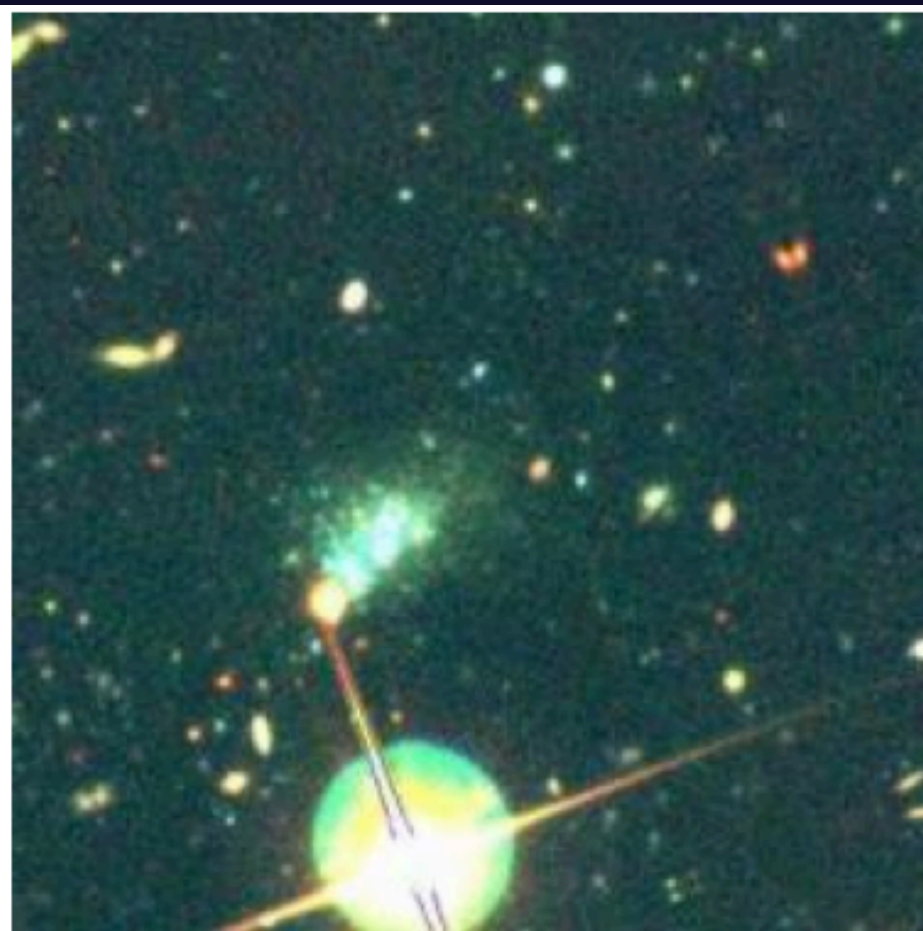
After layer unfreezing fine-tuning phase



Tried to find the best epoch where the validation loss was the minimum.



Some blue blobs
classified as
background or
galaxies.



Some background
images classified
as blue blobs



Overall Winner

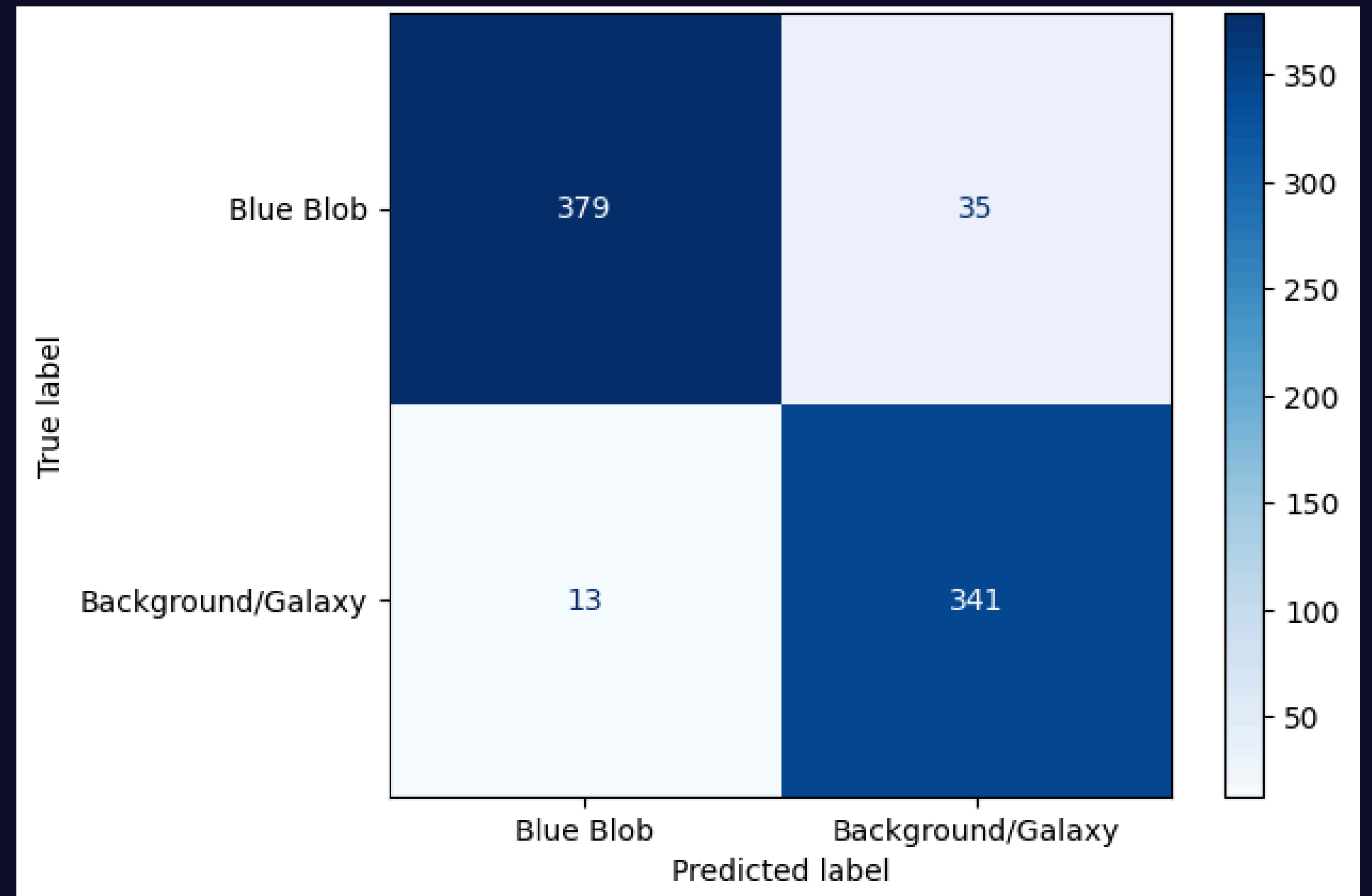
ResNet II

- The least amount of blue blobs are misclassified as galaxies.
- Also, pretty good background/galaxy recall.

Future + answering problem 3

I would not use this model for blue blob detection.

But I will use it to make a more efficient citizen science project where volunteers don't have to go through 100,000 cutouts of no value detections.



Thank you to Misha, Robert, and Arvind for advice on how to make the project better.